

Fig. 1.14 Integrable dynamics may be viewed as periodic or quasi-periodic motion on the surface of a torus (illustrated for a system with two degrees of freedom)

1.5.3 Equilibrium Points, and the Linearization Theorem

Molecular systems can be written in the form

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{f}(z). \tag{1.6}$$

An *equilibrium point* of such a system is a solution of $\boldsymbol{f}(z) = 0$.[4] An equilibrium point z^* corresponds to an equilibrium solution, since if we define a constant function $z(t) \equiv z^*$ then we have $\dot{z}(t) \equiv \boldsymbol{f}(z^*) = 0$. In some cases, it is possible to discuss the motion of solutions close to equilibrium points by means of *linearization*: the technique of replacing a nonlinear system by a linear one which approximates it locally.

We assume that $\boldsymbol{f}$ is continuously differentiable in the vicinity of the equilibrium point z^* and make use of the fact that $\boldsymbol{f}(z) \approx \boldsymbol{f}(z^*) + \boldsymbol{f}'(z^*)(z - z^*)$ for $\|z - z^*\|$ sufficiently small. Since $\boldsymbol{f}(z^*) = 0$ we have, defining $\delta z := z - z^*$,

$$\frac{\mathrm{d}\delta z}{\mathrm{d}t} \approx \boldsymbol{A}\delta z, \qquad \boldsymbol{A} = \boldsymbol{f}'(z^*).$$

The symbol $\approx$ is not very precise. In general, the approximation of solutions is only local in time meaning that solutions of the nonlinear system and the linear approximation may diverge. We refer to the system

$$\frac{\mathrm{d}\delta z}{\mathrm{d}t} = \boldsymbol{f}'(z^*)\delta z$$

as the *linearization* of the system (1.6) around z^*.

In case the equilibrium point is *hyperbolic*, meaning that the real parts of the eigenvalues of $\boldsymbol{A} = \boldsymbol{f}'(z^*)$ are nonzero, then one can infer that the solutions of the

[4]For readers conversant with concepts of statistical mechanics and in preparation for later discussions, we emphasize that the concept of *mechanical equilibrium* given in relation to the system of ordinary differential equation (1.6) is entirely different from the notion of *statistical* or *thermal equilibrium* of Chap. 6.